Cambridge (CIE) A Level Maths: Pure 1

Your notes

Binomial Expansion

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Binomial expansion

What is the binomial expansion?

$$
\begin{aligned}
(a+b)^n = {} & {}^nC_0(a)^n(b)^0 \\
& + {}^nC_1(a)^{n-1}(b)^1 \\
& + {}^nC_2(a)^{n-2}(b)^2 \\
& + {}^nC_3(a)^{n-3}(b)^3 \\
& + \dots \\
& + {}^nC_r(a)^{n-r}(b)^r \\
& + \dots \\
& + {}^nC_{n-1}(a)^1(b)^{n-1} \\
& + {}^nC_n(a)^0(b)^n
\end{aligned}
$$

$$
{}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} \qquad n \in \mathbb{N}
$$

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- Look at the pattern
  - Start at ${}^nC_0$, then ${}^nC_1$, ${}^nC_2$, etc
  - Powers of **a** start at **n** and decrease by 1
  - Powers of **b** start at 0 and increase by 1
- There are shortcuts but these hide the pattern
  - ${}^nC_0 = {}^nC_n = 1$
  - ${}^nC_1 = {}^nC_{n-1} = n$
  - ${}^nCr = {}^nC_{n-r}$
  - $(b)^0 = (a)^0 = 1$
- Use the shortcuts once familiar with the pattern
- ! means factorial

$$(a+b)^n = a^n + na^{n-1}b + {}^nC_2 a^{n-2}b^2 + \ldots + b^n$$

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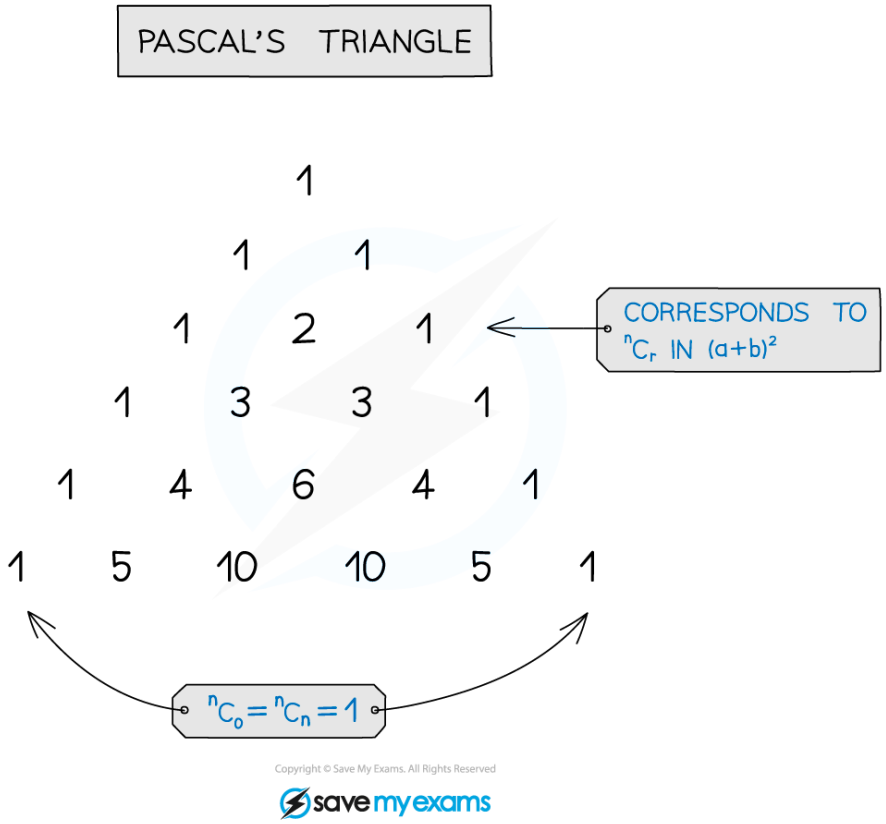
- This is given in the formula booklet

What is Pascal's triangle?



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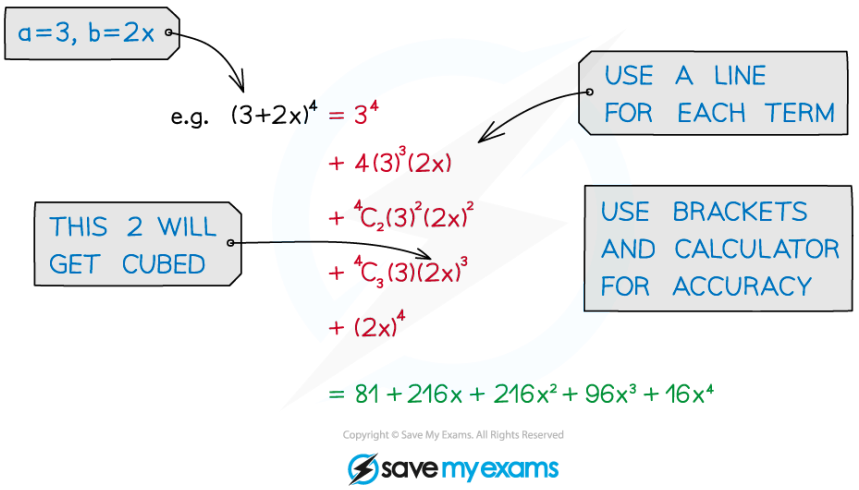
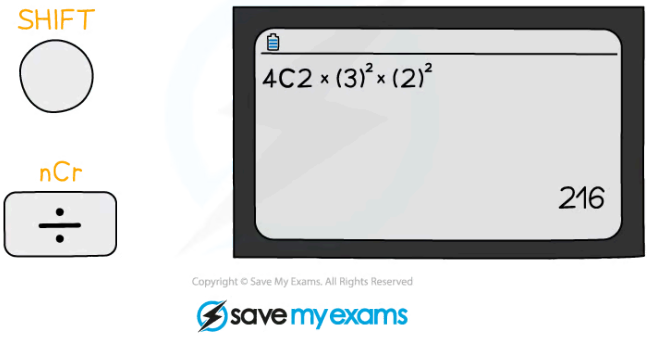
- Pascal's triangle is an alternative to ^nC_r
- It is useful for lower values of **n**
- For larger **n** it is slow and prone to arithmetic errors

How do I expand brackets with the binomial expansion?



- Use a line for each term to make things easier to read and follow
- Use brackets, particularly helpful when negatives involved
- Use a calculator for ${}^nC_r$



Worked Example

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Find the first four terms, in ascending powers of x, in the expansion of $(2-3x)^6$

USE SEPARATE LINES AND BRACKETS TO KEEP WORKING CLEAR

$$(2-3x)^6 \approx (2)^6 + 6(2)^5(-3x) + {}^6C_2(2)^4(-3x)^2 + {}^6C_3(2)^3(-3x)^3$$

TYPE ${}^6C_3 \times (2)^3 \times (-3)^3$ INTO CALCULATOR

$$= 64 - 576x + 2\,160x^2 - 4\,320x^3$$

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Applications of binomial expansion

What are binomial expansions used for?

- Expanding brackets, it is usual to be asked for either …
  - … the first few terms only

e.g. FIND, IN ASCENDING POWERS OF x, THE FIRST THREE TERMS OF $(5+2x)^9$

$$(5+2x)^9 \approx (5)^9 + 9(5)^8(2x) + {}^9C_2(5)^7(2x)^2$$

$$= 1\,953\,125 + 7\,031\,250x + 11\,250\,000x^2$$

APPROXIMATELY EQUAL TO AS 3 TERMS ONLY

ASCENDING POWERS OF x SO START WITH CONSTANT TERM (x^0)

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- or (the coefficient of) a particular term

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e.g. FIND THE COEFFICIENT OF x^6 IN THE EXPANSION OF $(3+2x)^8$

START HERE, FOR "x^6" "$(b)^6$" IS NEEDED

$^8C_6\,(3)^2(2x)^6$

TOTAL OF POWERS IS "n" WHICH HERE IS 8

MATCHES POWER OF "b"

$= 16\,128x^6$

SO THE COEFFICIENT OF x^6 IS 16 128

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- **Solving problems** in unknowns

e.g. THE COEFFICIENT OF THE x^3 TERM IN THE EXPANSION OF $(p+2x)^5$ IS 2000. FIND THE VALUE OF p, GIVEN $p>0$

ONLY NEED THE TERM IN x^3

$^5C_3\,(p)^2(2x)^3 = 80p^2x^3$

$80p^2 = 2000$

$p^2 = 25$

$\cancel{p=-5}$ $p=5$

SAY WHY THIS SOLUTION IS REJECTED

REJECT SINCE $p>0$

$p=5$

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